

VPA ICT — Year 7: Programming & Databases

Year 7: Programming Fundamentals & Database Concepts

Duration: 36 weeks | Language: Python 3.x | Database: Microsoft Access / SQLite

Unit 1: Python Programming (Weeks 1 – 18)

Week 1 — Introduction to Python

Keywords: Python Python (编程语言), IDE 集成开发环境 (jíchéng kāifā huánjìng), interpreter 解释器 (jiěshìqì), syntax 语法 (yǔfǎ), execute 执行 (zhíxíng)

Activity 1 (15 min): Set up Replit.com or IDLE. Run first program:

```
print("Hello, World!")
```

Activity 2 (15 min): Experiment with print statements. Print name, age, favorite color on separate lines.

Activity 3 (10 min): Debugging — Fix 3 broken print statements (missing quotes, wrong capitalization).

Week 2 — Variables & Data Types

Keywords: variable 变量 (biànlìang), string 字符串 (zìfú chuàn), integer 整数 (zhěngshù), float 浮点数 (fúdiǎn shù), assignment 赋值 (fùzhí)

Activity 1 (15 min): Create variables for personal information:

```
name = "Li Wei"
```

```
age = 12
```

```
height = 1.55
```

```
print(name, "is", age, "years old.")
```

Activity 2 (15 min): Type checking — Use type() function to identify data types.

Activity 3 (10 min): Variable naming rules — Why can't we start with a number? Why no spaces?

Week 3 — User Input

Keywords: input 输入 (shūrù), prompt 提示 (tíshì), string concatenation 字符串连接 (zìfú chuàn liánjiē), type casting 类型转换 (lèixíng zhuǎnhuàn)

Activity 1 (15 min): "Mad Libs" program — Ask for noun, verb, adjective; print silly story.

Activity 2 (15 min): Calculator v1 — Ask for 2 numbers, add them (remember int()!).

Activity 3 (10 min): Error handling — What happens if user types "five" instead of 5?

```
num1 = int(input("Enter first number: "))
```

```
num2 = int(input("Enter second number: "))
```

```
print("Sum:", num1 + num2)
```

Week 4 — Arithmetic Operations

Activity 1 (15 min): Build a BMI calculator: `weight / (height ** 2)`

Activity 2 (15 min): Temperature converter — Celsius to Fahrenheit and back.

Activity 3 (10 min): Math challenge — Calculate area of circle, triangle, rectangle with user input.

Week 5 — If/Else Statements

Keywords: `if` 如果 (`rúguǒ`), `else` 否则 (`fǒuzé`), `elif` 否则如果 (`fǒuzé rúguǒ`), `condition` 条件 (`tiáojiàn`), `comparison operator` 比较运算符 (`bǐjiào yùnsuàn fú`)

Activity 1 (15 min): Grade calculator — Input score, output grade (A/B/C/D/F).

Activity 2 (15 min): "Guess the Number" game — Computer picks 7; tells user if guess is too high/low.

Activity 3 (10 min): Login simulator — Check username and password.

```
score = int(input("Enter score: "))
```

```
if score >= 90:
```

```
    print("Grade A")
```

```
elif score >= 80:
```

```
    print("Grade B")
```

```
else:
```

```
    print("Keep trying!")
```

Week 6 — Loops: For & While

Keywords: `for loop` for循环 (`for xúnhuán`), `while loop` while循环 (`while xúnhuán`), `iterate` 迭代 (`diédài`), `range` 范围 (`fànwei`), `counter` 计数器 (`jìshùqì`)

Activity 1 (15 min): Multiplication table generator — `for i in range(1, 11): print(n, "x", i, "=", n*i)`

Activity 2 (15 min): Countdown timer using while loop.

Activity 3 (10 min): "When do I use for vs. while?" decision tree.

Week 7 — Lists

Keywords: `list` 列表 (`lièbiǎo`), `index` 索引 (`suǒyǐn`), `append` 添加 (`tiānjiā`), `remove` 删除 (`shānchú`), `length` 长度 (`chángdù`)

Activity 1 (15 min): Shopping list program — Add items, remove items, display list.

Activity 2 (15 min): "Top 5 Songs" — Create list, sort alphabetically, reverse order.

Activity 3 (10 min): List comprehension intro — Create list of squares: `[x**2 for x in range(1, 11)]`

Week 8 — Functions

Keywords: `function` 函数 (`hánshù`), `parameter` 参数 (`cānshù`), `return` 返回 (`fǎnhuí`), `call` 调用 (`diàoyòng`), `def` 定义 (`dìngyì`)

Activity 1 (15 min): Create `greet(name)` function. Call it 3 times with different names.

Activity 2 (15 min): Calculator functions — `add(a,b)`, `subtract(a,b)`, `multiply(a,b)`, `divide(a,b)`.

Activity 3 (10 min): Why functions? Discussion on DRY principle (Don't Repeat Yourself).

```
def calculate_area(length, width):
```

```
    return length * width
```

```
print(calculate_area(5, 3))
```

Week 9 — String Manipulation

Activity 1 (15 min): Password validator — Check length, uppercase, number.

Activity 2 (15 min): Text analyzer — Count words, characters, find specific words.

Activity 3 (10 min): Palindrome checker — Is "radar" the same forwards and backwards?

Week 10 — File Handling

Keywords: file 文件 (wénjiàn), open 打开 (dǎkāi), read 读取 (dúqǔ), write 写入 (xiěrù), close 关闭 (guānbì)

Activity 1 (15 min): Read a text file and count lines.

Activity 2 (15 min): Write a diary entry to a file with timestamp.

Activity 3 (10 min): Quiz program — Read questions from file, check answers.

Week 11 — Mini Project: Quiz Game

Project Requirements:

5+ questions stored in a list or file

Score tracking with variable

Feedback for correct/incorrect answers

Final score display with percentage

Option to play again

Week 12 — Python Assessment

Practical Coding Test (40 min):

Write a program that asks for user's name and age, then calculates year of birth.

Create a function that returns the largest of 3 numbers.

Write a loop that prints even numbers from 2 to 20.

Read a list of 5 scores and calculate the average.

Write an if/else program that checks if a password is strong (8+ chars, has number).

Unit 2: Database Concepts (Weeks 13 – 24)

Week 13 — Introduction to Databases

Keywords: database 数据库 (shùjùkù), table 表 (biǎo), record 记录 (jìlù), field 字段 (zìduàn), primary key 主键 (zhǔjiàn)

Activity 1 (15 min): "Database vs. Spreadsheet" comparison. When do we need a database?

Activity 2 (15 min): Physical database model — Index cards as records. Sort by different fields.

Activity 3 (10 min): Identify fields in a student record: StudentID, Name, Class, DOB, Email.

Week 14 — Creating Tables in Access

Activity 1 (20 min): Create a "Students" table with: ID (AutoNumber), FirstName (Text), LastName (Text), Class (Text), DOB (Date), Email (Text).

Activity 2 (15 min): Set data types correctly. What happens if you put text in a number field?

Activity 3 (5 min): Set StudentID as primary key.

Week 15 — Data Entry & Validation

Activity 1 (15 min): Enter 10 student records. Practice navigation.

Activity 2 (15 min): Add validation rules: Class must be "7A", "7B", or "7C"; DOB must be before 2015.

Activity 3 (10 min): Create input masks for phone numbers.

Week 16 — Sorting & Filtering

Keywords: sort 排序 (páixù), filter 筛选 (shāixuǎn), ascending 升序 (shēngxù), descending 降序 (jiàngù), criteria 条件 (tiáojiàn)

Activity 1 (15 min): Sort students by LastName alphabetically. Then by DOB (oldest first).

Activity 2 (15 min): Filter to show only 7A students. Then only students born in 2012.

Activity 3 (10 min): Combined sort and filter challenge.

Week 17 — Simple Queries (SELECT)

Keywords: query 查询 (cháxún), SQL 结构化查询语言 (jiégòuhuà cháxún yǔyán), SELECT 选择 (xuǎnzé), FROM 从 (cóng), WHERE 条件 (tiáojiàn)

Activity 1 (15 min): Query design view — Show all students in 7B.

Activity 2 (15 min): SQL view introduction:

SELECT FirstName, LastName FROM Students WHERE Class = "7B";

Activity 3 (10 min): Create 3 queries: all students with A in first name; all born after 2011; all in 7A or 7B.

Week 18 — Database Assessment

Practical Test (40 min):

Create a "Library" database with Books table (BookID, Title, Author, Genre, Year, Available)

Enter 8 records

Create a query showing only "Fantasy" books

Create a query showing books published after 2020

Sort books by Author (A-Z)

Standards Mapping

Unit	Standard	Description
Python	CSTA 2-AP-10	Use flowcharts to plan and follow multistep instructions
Python	CSTA 2-AP-11	Create clearly named variables representing different data types
Databases	CSTA 2-DA-09	Refine computational models based on data

